

Problem

Consider a graph G with edges E and vertices V , and subgraphs $G_1, \dots, G_k \subset G$. Let $G_i = (E_i, V_i)$ for these subgraphs. For now let us suppose the graph is unweighted and undirected. The subgraphs need not cover G and can overlap. Each player i knows G and G_i , but not the other player's graphs. (This is akin to a map of a city, where each player is a different regional taxi company.) The goal is, given some starting vertex $v \in V$, and some end vertex, $w \in V$ to:

1. Identify if it is possible to construct a path $v, v_1, \dots, v_l, w$ such that every edge on the path is in $\bigcup_{i=0}^k E_i$.
2. If such a path exists, find the minimum number of edges needed to get from v to w , while using only edges belonging to $\bigcup_{i=0}^k E_i$.

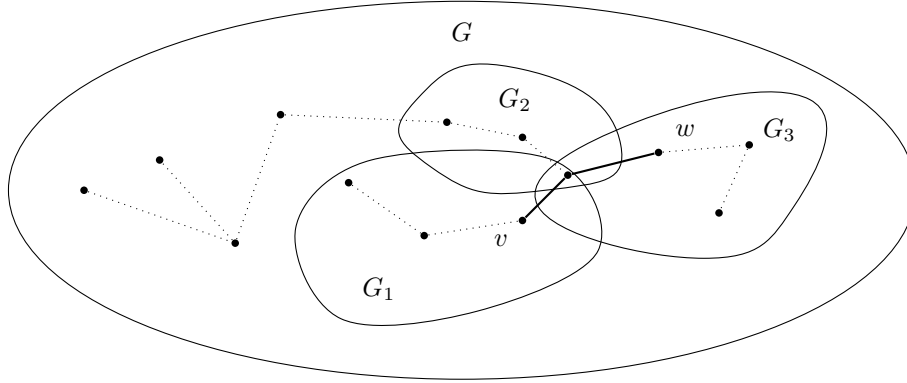


Figure 1: A valid path from v to w in bold. Note that G may contain nodes not belonging to any player.

Lower bound sketches and ideas

For item 1, we think this would require computing $\binom{k}{2}$ pairwise set disjointness in the worst case, to determine where the intersections of each G_i are, at which point each player could on their own determine the possible paths between each of the subgraphs that intersect with their own.

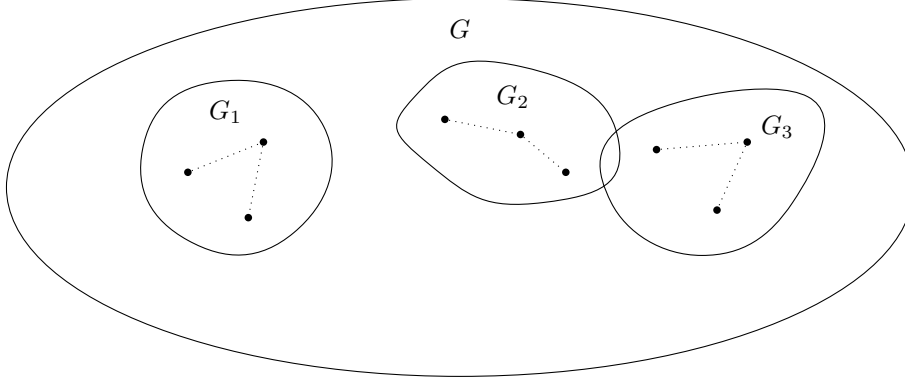


Figure 2: Worst case for item 1: Disjoint Subsets

For item 2, we would be required to compute $\binom{k}{2}$ pairwise intersections, since in order to minimize the path we would need to know all possible paths that could connect v and w . To find the shortest path, each player works backwards and sends the distances through their internal vertices to the players they intersect with, and each player will attempt to identify if a longer path can be dropped whenever they receive this list.